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ORIGINAL RESEARCH

Accessibility of clinical study reports supporting medicine approvals: a cross-sectional evaluation

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Abstract

Objectives: Clinical study reports (CSRs) are highly detailed documents that play a pivotal role in medicine approval processes. Though not historically publicly available, in recent years, major entities including the European Medicines Agency (EMA), Health Canada, and the US Food and Drug Administration (FDA) have highlighted the importance of CSR accessibility. The primary objective herein was to determine the proportion of CSRs that support medicine approvals available for public download as well as the proportion eligible for independent researcher request via the study sponsor.

Study Design and Setting: This cross-sectional study examined the accessibility of CSRs from industry-sponsored clinical trials whose results were reported in the FDA-authorized drug labels of the top 30 highest-revenue medicines of 2021. We determined (1) whether the CSRs were available for download from a public repository, and (2) whether the CSRs were eligible for request by independent researchers based on trial sponsors' data sharing policies.

Results: There were 316 industry-sponsored clinical trials with results presented in the FDA-authorized drug labels of the 30 sampled medicines. Of these trials, CSRs were available for public download from 70 (22%), with 37 available at EMA and 40 at Health Canada repositories. While pharmaceutical company platforms offered no direct downloads of CSRs, sponsors confirmed that CSRs from 183 (58%) of the 316 clinical trials were eligible for independent researcher request via the submission of a research proposal. Overall, 218 (69%) of the sampled clinical trials had CSRs available for public download and/or were eligible for request from the trial sponsor.

Conclusion: CSRs were available from 69% of the clinical trials supporting regulatory approval of the 30 medicines sampled. However, only 22% of the CSRs were directly downloadable from regulatory agencies, the remaining required a formal application process to request access to the CSR from the study sponsor. Crown Copyright © 2024 Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: Clinical trial transparency; Clinical study reports; Blockbuster medicines; Industry sponsored trials; Data sharing; Data transparency; Pharmaceutical industry; Regulatory decision transparency

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What is new?

Key findings

- Only 22% of clinical trials for top-revenue medicines had publicly accessible clinical study reports (CSRs).
- Pharmaceutical companies did not offer open access to CSRs, highlighting a need for industry transparency.

What this adds to what was known?

- This study provides a comprehensive assessment of CSR accessibility, revealing challenges and emphasizing transparency gaps.

What is the implication and what should change now?

- Open access models are crucial for improving CSR availability, fostering trust, and supporting informed healthcare decisions.

1. Introduction

Clinical study reports (CSRs) are comprehensive documents, often spanning hundreds of pages, that detail the conduct and results of clinical trials [1−3]. Their in-depth nature means that they can often offer alternate insights compared to more condensed journal publications or result synopses—albeit both these documents obviously also still have important uses [4]. Regardless, the importance of CSR transparency is highlighted by their pivotal role in medicine approvals, where pharmaceutical companies submit CSRs to regulatory bodies to present detailed evidence on the safety and efficacy of new medicines [1−3]. Despite their importance for enabling independent comprehensive medical product evaluations, CSRs have historically been inaccessible in the public domain [1−3,5,6].

Over the past decade, prominent bodies like Cochrane, European Federation of Statisticians in the Pharmaceutical Industry, Institute of Medicine (now the National Academy of Medicine), International Coalition of Medicines Regulatory Authorities, and the World Health Organization, among others, have underscored the importance of public access to CSRs for facilitating scientific research, trust in regulatory decisions, as well as comprehensive medical product evaluations [2,5−14]. This becomes particularly important with regard to the adverse event profiles of newer medicines and for understanding subgroup effects. Responding to this global call, both the European Medicines Agency (EMA-effective 2015) and Health Canada (effective 2019) have adopted policies fostering CSR sharing [1,9,15−19]. The US Food and Drug Administration (FDA) has also issued statements supporting CSR sharing, albeit it appears to favor sponsor-led voluntary models to overcome efficiency issues associated with multiple regulators sharing the same documents [1,20].

In line with the above progress, two prominent CSR sharing approaches have been developed and promoted [2,5,6,10−14]. The EMA and Health Canada have recently established online repositories that enable direct public downloading of CSRs which have been submitted to them in medicine approval applications [15−18]. Simultaneously, many pharmaceutical companies have implemented data sharing policies to enable independent researchers to request access to CSRs [21,22]. Albeit, CSR access via these sponsor-led processes typically requires a formal application involving submission and review of a research proposal [2,7].

Despite the promotion of both CSR sharing approaches, the effectiveness of the recent policy updates by the EMA and Health Canada in facilitating CSR availability remains to be determined. The primary objective herein was to determine the proportion of CSRs available for public download, as well as the proportion eligible for independent researcher request via the study sponsor, for CSRs that support medicine approvals.

2. Materials and methods

2.1. Sample

This was a quality assessment, cross-sectional analysis study. It was estimated that a sample size of approximately 300 clinical trials would enable estimation of the proportion of CSRs available with sufficient precision (confidence interval ± 5%) to yield meaningful insights. Clinical trials supporting the registration of the highest-revenue medicines were sampled because these medicines are typically both commonly used and relatively recently registered. Based on pilot evaluations of FDA-authorized drug labels, it was estimated that an evaluation of the top 30 highest-revenue medicines would provide the required sample size.

Commencing in September 2022, a list of the top 30 highest-revenue medicines of 2021 was compiled from ‘Drug Discovery & Development’ and ‘S&P Global Market Intelligence’ data [23,24]. For each medicine, it was confirmed that they were FDA, EMA, and Health Canada approved. The most current FDA-authorized drug label for each medicine was then retrieved from the FDA Drug Database [25,26]. From these drug labels, a list of identifiers for clinical trials with results summarized in the labels was created. The time since the trial results were added to the drug label was noted by comparison to historical versions of the drug label. Additionally, for each trial information on the National Clinical Trial (NCT) number (or other identification if no NCT existed), CSR sponsor and trial phase were collected—this information was gathered from ClinicalTrials.gov, the EU Clinical Trials Register, or

journal publications for the trial which had been identified via PubMed and Google searches.

2.2. Determination of CSR availability

The primary objective of the study was to evaluate whether the full CSR for each clinical trial listed in the drug's label was publicly available for download. For each trial, two researchers were responsible for independently searching clinicaldata.ema.europa.eu, clinical-information.canada.ca, ClinicalTrials.gov, and medicines owner webpages to locate an accessible and directly downloadable CSR (ie, was a CSR available for download without gatekeeping and a potentially lengthy request process). In instances of differing findings between the two researchers, a direct comparison and discussion were undertaken to resolve any discrepancies and confirm the accessibility status of each CSR.

It is also recognized that many pharmaceutical companies have data sharing policies to facilitate independent researchers to request access to CSRs[21,22]. As a secondary objective, this study evaluated the eligibility of independent researchers to request access to CSRs from the trial sponsors for each clinical trial listed in the evaluated drug labels. This involved assessing the eligibility to request access to CSRs identified as available for download from other sources. The eligibility of each CSR for sharing was confirmed by either identifying a public listing of the CSR as eligible for request or by receiving a positive response to a standardized inquiry sent to the CSR sponsor (see Supplementary File 1). Ineligibility for CSR sharing was obtained through a negative response to the inquiry (ie, written confirmation from the sponsor that the CSR was ineligible for sharing). If a clinical trial was indicated as ineligible for CSR sharing, details of the reason(s) for ineligibility and when the CSR would become eligible were requested. If no response to the inquiry was received, reminders were sent at approximately 30-day intervals. If no clear response on the eligibility to request access to a specific CSR had been received by 180 days, then the CSR was deemed ineligible for access.

All statistical analyses were performed using R 4.2.2 [27]. 95% confidence intervals on observed proportions were calculated using a normal approximation of the binomial distribution. Chi-square tests, with statistical significance set at $P < 0.05$, were used to examine potential differences in CSR availability based on clinical trial phase and time since the results were added to the drug label.

3. Results

3.1. Sample

The FDA-authorized drug labels for the top 30 highest-revenue medicines of 2021 included the results of 316 industry-sponsored clinical trials. All evaluated medicines were FDA, EMA, and Health Canada approved. The sample included four vaccines (39 trials), four biologic anticancer medicines (76 trials), six non-biologic anticancer medicines (38 trials), nine biologic medicines for noncancer diseases (102 trials), and seven non-biologic medicines for noncancer diseases (61 trials). Of the 316 trials, 261 (83%) were phase 3 trials, while the results of 145 trials were added to the drug label within the past 5 years, 95 were added between 5 and 10 years ago, and 76 were added more than 10 years ago. Across the 316 trials, there were 20 unique pharmaceutical company sponsors, 16 (80%) of which were within the top 20 global revenue pharmaceutical companies of 2021 [28].

Supplementary Table 1 summarizes the characteristics, public availability of CSRs, and eligibility for independent researchers to request CSRs from the 30 sampled medicines. Fig. 1 presents a flow diagram summarizing both the public availability of CSRs and their eligibility for independent researcher request from the 316 clinical trials evaluated in this study. The raw data collected in this study can be found in Supplementary File 2.

3.2. Availability of CSRs for public download

In the first round of documenting the availability of CSRs for public download, there were just two instances of disagreement between the evaluators, which were promptly reconciled.

Of the 316 clinical trials evaluated, 70 (22%; 95% CI 18−27%) were identified as having CSRs publicly available for download. This included 37 CSRs available from clinicaldata.ema.europa.eu and 40 from clinical-information.canada.ca. No CSRs were publicly available for download from ClinicalTrials.gov or from pharmaceutical company webpages. Fig. 2A presents the percentage of CSRs publicly available for download for each medicine sampled.

The availability of CSRs for public downloads varied with the time since the clinical trial results were added to the drug label: 37 of 145 (26%) for trials less than 5 years since added, 26 of 95 (27%) for those added between 5 and 10 years, and 7 of 76 (9%) for those added over 10 years ago ($P < 0.01$). A similar pattern was observed with trial phase: 49 of 261 (19%) for phase 3 trials and 21 of 55 (38%) for non-phase 3 trials ($P < 0.01$).

3.3. CSR eligibility for request

In the first round of documenting the eligibility of CSRs for requests, there were no instances of disagreement between the evaluators.

Out of the 20 pharmaceutical companies sponsoring the clinical trials sampled included in our study, 16 (80%) had transparency policies that were publicly accessible. These public policies clearly outline the process whereby independent researchers can request access to CSRs from

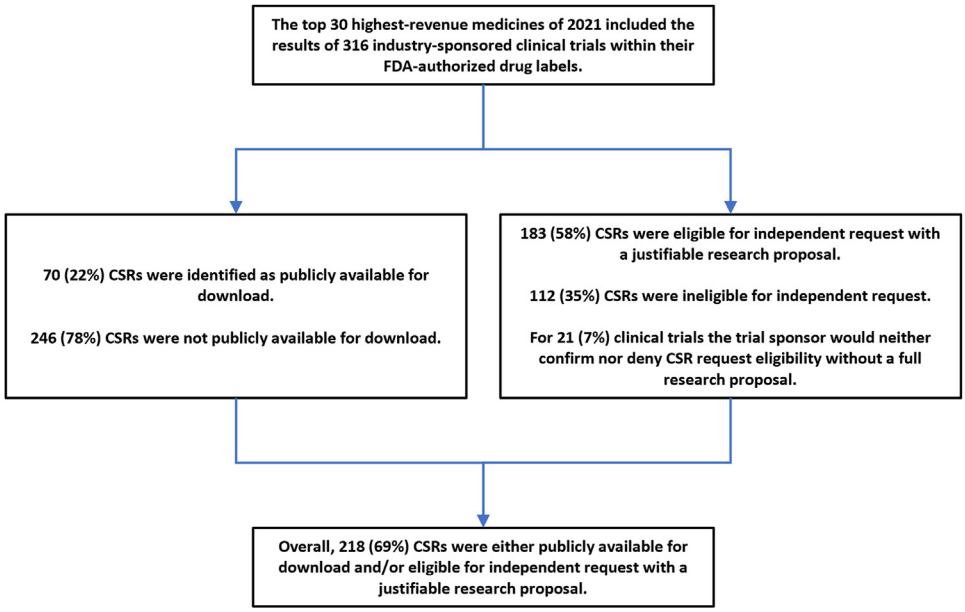


Fig. 1. Flow diagram summarizing both the public availability of CSRs and their eligibility for independent researcher request for the 316 industry-sponsored clinical trials with results presented in the FDA-authorized drug labels of 2021's top 30 highest-revenue medicines. CSRs, clinical study reports; FDA, Food and Drug Administration. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

clinical trials eligible for sharing through the submission of a detailed research proposal. Among these 16, 9 shared CSRs via Vivli [29], 2 through clinicalstudydatarequest.com [30], 1 through the Yale University Open Data Access (YODA) Project [31], and 4 through internal company processes.

Of the 316 clinical trials evaluated, 183 (58%; 95% CI 52−63%) had CSRs confirmed eligible for independent researcher request, 112 (35%; 95% CI 30−41%) were confirmed as ineligible for request, and for 21 (7%; 95% CI 4−9%), the trial sponsor would neither confirm nor deny CSR eligibility for request without a full research proposal. Fig. 2B presents the percentage of clinical trial CSRs confirmed eligible for independent request for each medicine sampled. The eligibility to request CSRs varied with the time since the clinical trial results were added to the drug label: 71 of 145 (49%) for trials less than 5 years since added, 63 of 95 (66%) for those added between 5 and 10 years, and 49 of 76 (64%) for those added over 10 years ago ($P = 0.01$). In contrast, eligibility to request CSRs was not observed to differ according to trial phase: 156 of 261 (60%) for phase 3 trials and 27 of 55 (49%) for non-phase 3 trials ($P = 0.14$).

Table 1 presents a breakdown of reasons reported by the trial sponsors for each of the 133 (42%) clinical trials to which eligibility to request a CSR could not be confirmed. The three most common reasons provided for confirmed ineligibility to request were the company does not share CSRs (49 trials, 16%), the trial is ongoing (24 trials, 8%), and the clinical trial initiation or completion date predates the companies CSR sharing policy (11 trials, 3%).

3.4. Overall accessibility of CSRs

Of the 316 clinical trials evaluated, it was confirmed that 218 (69%; 95% CI 64−74%) had CSRs available for public download and/or were eligible for independent researcher request via the submission of a research proposal (Fig. 1).

4. Discussion

In this cross-sectional assessment of the accessibility of CSRs from 316 industry-sponsored clinical trials with results presented in the FDA-authorized drug labels of the top 30 highest-revenue medicines of 2021, only 70 (22%) had a CSR publicly available for download. This included 37 CSRs made available by the EMA and 40 by Health Canada. No CSRs were freely available for public download via company-led initiatives, as any CSRs available directly from trial sponsors required a research proposal to be submitted to secure access. However, overall, 218 (69%) of the sampled CSRs were available when combining both the regulatory processes with pharmaceutical company options for accessing CSRs.

The extent to which contemporary regulatory agency policies effectively facilitate public access to CSRs remains largely unknown, as it has not been systematically assessed. Considering the significant policy changes over the past decade—with the EMA's policy effective from 2015 and Health Canada's from 2019—benchmarking the current state of these initiatives is important to ensure they are meeting their intended outcomes. Our findings show that

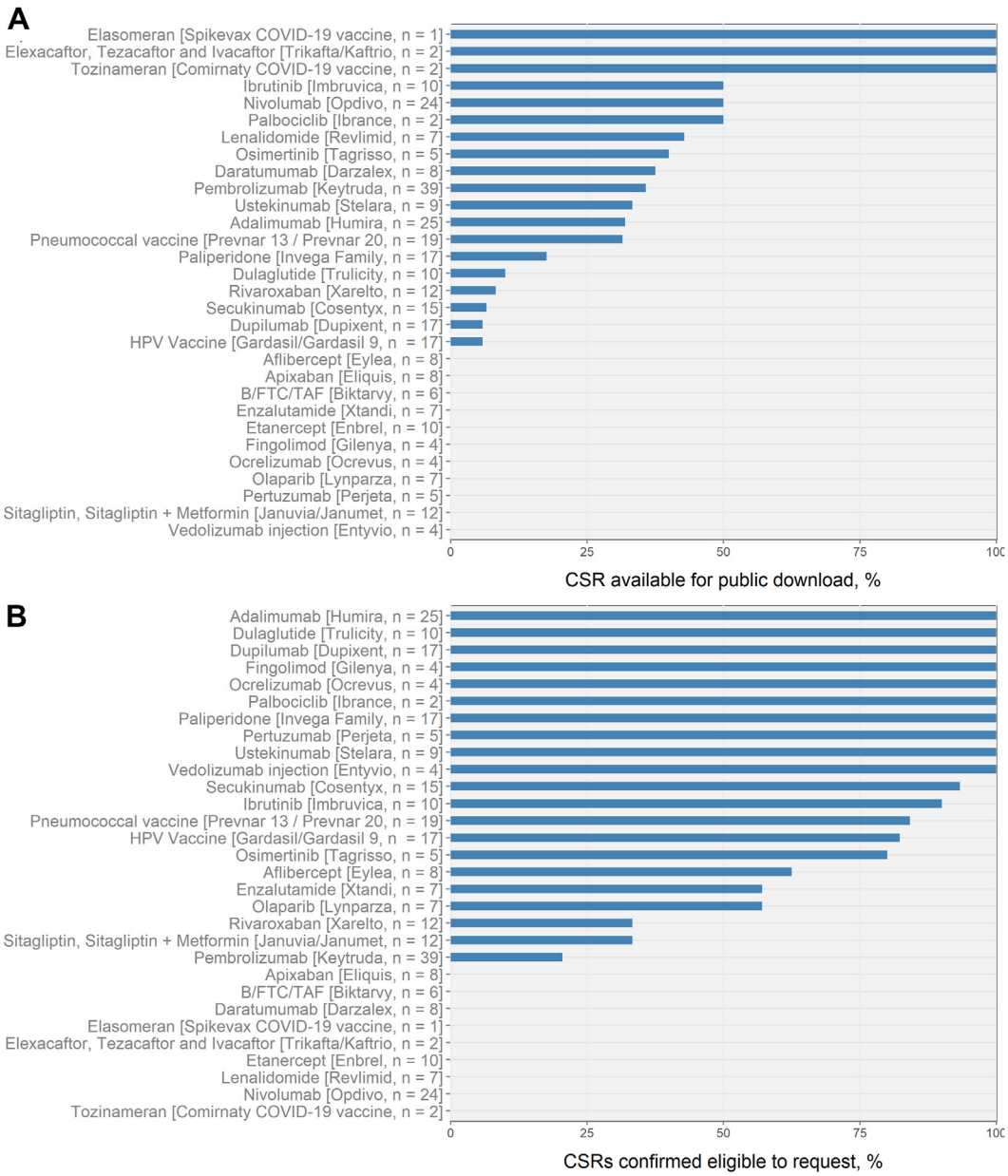


Fig. 2. (A) displays the percentage of CSRs that were publicly available for download, and (B) displays the percentage of CSRs eligible for independent research request for the industry-sponsored clinical trials with results presented in the FDA-authorized drug labels of 2021's top 30 highest-revenue medicines. CSRs, clinical study reports; FDA, Food and Drug Administration. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

22% of the sampled trials had CSRs available for download across the repositories led by EMA and Health Canada, underscoring both considerable progress over the past 10 years and a large opportunity for further improvement in public accessibility to CSRs that support medicine approvals.

There appear to be two primary factors that may have contributed to the relatively small proportion of CSRs publicly available via regulatory agencies. Firstly, Health Canada has only published a limited number of CSRs that predate their commitment to transparency [17]. Health Canada has expressed intentions to make CSRs predating 2019 available; however, it remains to be seen how this will evolve over time. Secondly, the EMA halted its CSR sharing initiative in December 2018 [4], citing operational difficulties (Supplementary File 3). It can be hoped that the process will resume, but presently the halt accentuates the value of Health Canada's 2019 policy as a critical measure upon which public CSR access moving forward depends. The EMA halt also aligns with the FDA's 2020 assertions that, while supporting CSR sharing, an international library backed by sponsors might be necessary to bypass logistical

Table 1. Breakdown of clinical trials for which eligibility to request CSR access was not confirmed

Reasons provided	Total assessed = 316
Clinical trials confirmed as ineligible for CSR request	112 (35%)
Sponsor does not share CSRs	49 (16%)
Clinical trial is ongoing	24 (8%)
Clinical trial initiation or completion date predates CSR sharing policy	11 (3%)
Not in scope for sharing per policy	9 (3%)
Phase 1 clinical trials are out of scope	7 (2%)
CSR under embargo, as not yet 18 mo after clinical trial completion	6 (2%)
Ongoing regulatory activities	3 (1%)
Codevelopment/contractual constraints	3 (1%)
Clinical trials unable to be confirmed eligible/ineligible for CSR request	21 (7%)
Full research proposal required to assess eligibility to share a CSR	21 (7%)

Data specified as number of trials (% of the total number of trials assessed).

Abbreviation: CSRs, clinical study reports.

and efficiency challenges that regulators face in preparing and sharing these documents [20]. However, achieving such a CSR sharing ecosystem undoubtedly requires precise regulatory action; notably, our study revealed that none of the 20 trial sponsors in our sample currently share CSRs through an open access (publicly available download) model. Encouragingly, functionalities already exist for uploading supporting documents, including CSRs, on clinical trial registration websites (eg, clinicaltrials.gov) [32,33]. We assert that these resources represent a promising pathway for comprehensive voluntary disclosure of CSRs.

Alongside open access models for CSR sharing, many pharmaceutical companies have implemented data sharing policies that allow independent researchers to request access to CSRs via a formal application process involving submission and review of a research proposal [21,22]. Notably, request-based processes present challenges to timely access to CSRs, given the need for a detailed research proposal and statistical analysis plan [29]. The process of developing proposals that satisfy the stringent criteria of each sponsor can span several weeks, and even after submission, it can take months before access to the CSRs can be negotiated [29,34]. In this study, 58% of the sampled clinical trials were confirmed by the trial sponsor to be eligible for CSR sharing. Thus, currently, a CSR is more likely to be accessible via the trial sponsor than via a regulatory agency. To the best of the authors' knowledge, there are no prior studies that quantify CSR accessibility from pharmaceutical companies. The 58% accessibility observed likely represents a marked improvement over what was possible 10 years ago. Nonetheless, 42% of CSRs were not confirmed to be available and hence there is still room for improvement. Our findings indicate that 16 (80%) of the 20 pharmaceutical companies that sponsored clinical trials in our sample had processes for independent researchers to request access to CSRs. This aligns with previous reports that between 80 and 96% of major pharmaceutical companies have data sharing policies which enable CSR access [21,35].

The rise in trial preregistration is a significant stride toward improving transparency in clinical research [36,37]. We propose that the next step in transparency be the broad dissemination of CSRs. In line with the International Committee of Medical Journal Editors acknowledgment of the crucial role preregistration plays in promoting transparency and mitigating publication bias [38], we advocate for a similar emphasis on CSR sharing. This strategy offers a more thorough insight into trial methodologies and results, going beyond the constraints of traditional publications. The encouraging trend in trial preregistration paves the way for potential progress through systematic CSR sharing.

Rigorous systematic reviews of clinical trial data are pivotal sources of synthesized evidence for interventions [6]. Concurrently, the rapid emergence of advanced large language models herald's capabilities to collate, summarize, and analyze vast amounts of literature in real time. Significantly, these capabilities could revolutionize the utility of CSRs in clinical practice. They have the potential to empower clinicians to easily explore complex patterns and nuances in adverse events and outcomes information across multiple extensive documents, thereby enabling informed, personalized patient care. However, this potential can only be realized in an environment where access to CSRs is unbiased and openly available. Cochrane emphasizes that the presence of retrieval bias can markedly compromise the validity of systematic reviews and meta-analyses [2,39]. Our study underscores this issue, indicating many CSR documents are currently unavailable, thus restricting their contribution to the body of post-approval evidence. Furthermore, previous research emphasizes that when CSRs are made available, the extent of redactions needs to be appropriate and, equally, consistent with informed consenting procedures [40–43].

This study sampled the clinical trials supporting the highest revenue-generating medicines on the basis that typically these medicines were both commonly used and included more recently registered medicines. The accessibility of CSRs from lower revenue or older medicines may differ. Illustrating this point, both Health Canada and the EMA, in addition to existing policies, specifically committed to rapid transparency for COVID-19 vaccines [15,44], resulting in all three clinical trials listed in the drug labels of the two COVID-19 vaccines in our study sample having CSRs available for download at both repositories. Furthermore, the ability to request CSRs from smaller companies likely differs [21,34,45]. Despite these inherent limitations, based on a sizable sample and a comprehensive approach, this research provides crucial insights into the current CSR sharing ecosystem. Future studies into CSR

accessibility should assess medicines exclusively approved by the EMA and Health Canada, extend coverage to non-high revenue medicines, and systematically explore CSRs for both nonindustry-sponsored trials and medicines that failed to receive FDA approval.

In conclusion, of the 316 clinical trials with results presented in the drug labels of the top 30 highest-revenue medicines of 2021, the CSRs were confirmed accessible from 218 (69%) clinical trials. However, only for 70 (22%) trials were the CSRs available for public download, with the remainder eligible for request from the trial sponsor via a formal application process which typically required submission and review of a research proposal. None of the pharmaceutical companies in our sample led initiatives making their CSRs openly available. CSR sharing is an important mechanism to bolster public trust in regulatory processes and stimulate further scientific discovery [2,6,15,17], and this study provides a benchmark to assess the success of initiatives for further improving CSR availability.

Role of the funder/sponsor

The funding organizations had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

CRediT authorship contribution statement

Ashley M. Hopkins: Writing − review & editing, Writing − original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Natansh D. Modi:** Writing − review & editing, Writing − original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Frank W. Rockhold:** Writing − review & editing, Writing − original draft, Methodology. **Tammy Hoffmann:** Writing − review & editing, Writing − original draft, Methodology. **Bradley D. Menz:** Writing − review & editing, Writing − original draft, Software, Data curation. **Areti-Angeliki Veroniki:** Writing − review & editing, Writing − original draft, Methodology. **Ross A. McKinnon:** Writing − review & editing, Writing − original draft, Methodology, Funding acquisition. **Andrew Rowland:** Writing − review & editing, Writing − original draft, Methodology, Investigation, Funding acquisition. **Sandra M. Swain:** Writing − review & editing, Writing − original draft, Methodology, Investigation. **Joseph S. Ross:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Investigation. **Michael J. Sorich:** Writing − review & editing, Writing − original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Data curation.

Data availability

The raw data collected in this study can be found in Supplementary File 2.

Declaration of competing interest

A/Prof Rowland and Prof Sorich are recipients of investigator-initiated funding for research outside the scope of the current study from AstraZeneca, Boehringer Ingelheim, Pfizer, and Takeda. A/Prof Rowland is a recipient of speaker fees from Boehringer Ingelheim and Genentech. Professor Swain reports grant to institution from Genentech-Roche and paid non-promotional speaking, consulting, and in-kind manuscript writing from Genentech−Roche. Professor Swain also reports grants to institution from Kailos Genetics. Professor Swain reports consulting fees paid from Athenex, Sanofi, AstraZeneca, Daiichi Sankyo, Lilly Pharmaceuticals, Merck, Molecular Templates, Biotheranostics, Natera, and Exact Sciences. Professor Swain reports support for attending meetings and travel from Sanofi, Daichi Sankyo, and Roche/Genentech. Third party in-kind manuscript writing from AstraZeneca. Dr Ross currently receives research support through Yale University from Johnson and Johnson to develop methods of clinical trial data sharing, from the Food and Drug Administration for the Yale-Mayo Clinic Center for Excellence in Regulatory Science and Innovation (CERSI) program (U01FD005938), from the Agency for Healthcare Research and Quality (R01HS022882), and from Arnold Ventures; formerly received research support from the Medical Device Innovation Consortium as part of the National Evaluation System for Health Technology (NEST) and from the National Heart, Lung and Blood Institute of the National Institutes of Health (NIH) (R01HS025164, R01HL144644); and in addition, Dr Ross was an expert witness at the request of Relator's attorneys, the Greene Law Firm, in a qui tam suit alleging violations of the False Claims Act and Anti-Kickback Statute against Biogen Inc. that was settled September 2022. Frank Rockhold receives research support from NIH, PCORI, BMS, AZ, American Regent, Gates Foundation, Eidos; Consulting for Janssen, Clover, Doctor Evidence and Intercept; DSMB for Lilly, AZ, Merck, Gilead, Novartis, Icosavax, Sanofi, UCB, Amgen, Biogen, BMS, pulmocide, Alkermes, Diurnal; nonprofit chairman of the board (unpaid) for Frontier Science Foundation; stock/stock options from GSK, Clover, Athira, Doctor Evidence, DataVant, Spencer Health Solutions, Adaptic Health. Dr Areti Angeliki Veroniki is a

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2024.111263.

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